

Odd-Rank Loci and Box-Induced Symmetries of Centered Sudoku Operators

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Abstract

We study Sudoku grids as a constrained class of Latin squares through the centered linear operator $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2}J_n$, where $\gamma \in S_n$ is a symbol relabeling and J_n is the all-ones matrix. Restricting to the standard hyperplane $V_{\text{std},n} = \{x \in \mathbb{Q}^n : \mathbf{1}^\top x = 0\}$, we prove four results. (a) A *Box Band Lemma*: $P_{\text{rb}} E_{\gamma,L} P_{\text{cs}}^\top = 0$, an exact linear identity that fails for generic Latin squares and isolates the box constraint at the linear-algebraic level. (b) The odd-rank locus $\Gamma_L^* = \{\gamma : \text{rank } E_{\gamma,L} = n - 2\}$ is structured: every odd-rank kernel certificate lifts to $V_{\text{std},n}$ and lies in the root lattice A_{n-1} , and at $n = 7$ the relevant subgroup of $B_3 < S_7$ is isomorphic to D_6 . (c) At $n = 9$ on the base M_{19} , the left/right Hessian set-stabilizers of the Hessian-class subset $\Gamma_{M_{19}}^*$ (of size 72) satisfy $K_L \cong K_R \cong 3^2:Q_8 = \text{SmallGroup}(72, 41)$, with an explicit single-coset structure $\Gamma_{M_{19}}^* = K_L \cdot \gamma_0 = \gamma_0 \cdot K_R$ of size 72, with $K_L = \gamma_0 K_R \gamma_0^{-1}$ inside the ambient coordinate stabilizer $\text{Stab}_{S_9}(u_9^*) \cong S_3 \times S_6$ of order 4320. (d) This structure is not explained by Latin-square autotopisms: $|\text{Aut}(L_{M_{19}})| = 1$, so the underlying Latin square is rigid. We complement these results by an analytic identification of the stabilizer dichotomy at $n \in \{7, 9, 11\}$, exhaustive computational verification of the corresponding orbit/stabilizer formulas for $5 \leq n \leq 13$, and a certified reproducibility package whose contents are pinned by a SHA-256 manifest.

Keywords. Sudoku, Latin squares, centered operators, odd-rank locus, A_{n-1} root lattice, Hessian group, autotopism.

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1 Introduction

1.1 From Latin squares to Sudoku via centered operators

Let L be a Latin square of order n on the symbol set $\{1, \dots, n\}$. The *centered operator* associated with a symbol relabeling $\gamma \in S_n$ is

$$E_{\gamma,L} := \gamma(L) - \frac{n+1}{2}J_n, \quad (1)$$

where $\gamma(L)$ is the entrywise application of γ and J_n is the $n \times n$ all-ones matrix. The operator $E_{\gamma,L}$ has constant row and column sums equal to zero, hence restricts to an endomorphism of the standard hyperplane $V_{\text{std},n} = \{x \in \mathbb{Q}^n : \mathbf{1}^\top x = 0\}$.

Three observations motivate the present paper. First, the centering in (1) is the same as in the sibling determinant-divisibility paper [2]; this paper studies the *rank* of $E_{\gamma,L}$, not its determinant. Second, the rank is an integer-valued invariant of the pair (L, γ) which is generically equal to $n - 1$ on $V_{\text{std},n}$ but can drop to $n - 2$ on a sparse, structured subset of S_n . Third, when L is *Sudoku* (i.e. also satisfies the $\sqrt{n} \times \sqrt{n}$ box constraint at $n = 9$), the rank-drop locus and its kernel certificates exhibit additional algebraic structure that does not arise from generic Latin squares.

1.2 Why box constraints matter

The box constraint is usually a combinatorial requirement: every $\sqrt{n} \times \sqrt{n}$ block of a Sudoku grid contains the symbols $\{1, \dots, n\}$. We promote it to an exact linear identity. Let $B_{\text{rb}}, B_{\text{cs}} \in \{0, 1\}^{n \times \sqrt{n}}$ be the row-band and column-stack incidence matrices, and let $P_{\text{rb}} := \frac{1}{\sqrt{n}} B_{\text{rb}}^\top$, $P_{\text{cs}} := \frac{1}{\sqrt{n}} B_{\text{cs}}^\top \in \mathbb{Q}^{\sqrt{n} \times n}$ be the corresponding band-mean and stack-mean projectors (defined precisely in §3.1). Then for every Sudoku grid L and every $\gamma \in S_n$,

$$P_{\text{rb}} E_{\gamma, L} P_{\text{cs}}^\top = 0. \quad (2)$$

This identity *fails* for generic Latin squares; that failure isolates the box contribution at the linear-algebraic level. The full formal statement of (2) is Theorem 3.1 below.

1.3 Main results

The four main contributions of the paper are stated informally below; each appears in formal form exactly once in the dedicated section indicated.

- (a) **Box Band Lemma (§3).** The identity $P_{\text{rb}} E_{\gamma, L} P_{\text{cs}}^\top = 0$ is an exact, Sudoku-specific linear identity. It fails for generic Latin squares. (*Theorem 3.1, certified.*)
- (b) **Odd-rank certificates lie in A_{n-1} (§6, §7).** For every relabeling γ producing odd rank $n - 2$ on $V_{\text{std}, n}$, the kernel certificate lifts to a vector in $V_{\text{std}, n}$ that lies in the root lattice $A_{n-1} = V_{\text{std}, n} \cap \mathbb{Z}^n$. At $n = 9$, the distinguished sparse-low template orbits M and O_6 saturate A_8 : $\Lambda_M = \Lambda_{O_6} = A_8$. (*Theorem 6.1; Theorem 7.1.*)
- (c) **Hessian symmetry of M_{19} (§8).** On the Sudoku base M_{19} , the left/right set-stabilizers of the Hessian-class subset $\Gamma_{M_{19}}^*$ satisfy $K_L \cong K_R \cong 3^2:Q_8 = \text{SmallGroup}(72, 41)$, and the Hessian-class subset $\Gamma_{M_{19}}^*$ (of size 72, sitting inside the full rank-locus $\Gamma_{\text{rank}}(M_{19})$ of size 100) form a single K_L -coset $\Gamma_{M_{19}}^* = K_L \gamma_0 = \gamma_0 K_R$ with $K_L = \gamma_0 K_R \gamma_0^{-1}$. The number $4320 = 3! \cdot 6!$ is the order of the ambient coordinate stabilizer $\text{Stab}_{S_9}(u_9^*) \cong S_3 \times S_6$, not the size of $\Gamma_{M_{19}}^*$. (*Theorem 8.5; Theorem 8.4.*)
- (d) **Sudoku-specificity beyond autotopisms (§10).** The Latin square underlying M_{19} is rigid as a Latin square: $|\text{Aut}(L_{M_{19}})| = 1$. Therefore K_L is *not* the image of any autotopism subgroup. The Hessian structure is genuinely Sudoku-grid / box / A_8 -induced. (*Theorem 10.1; Corollary 10.2.*)

We complement (a)–(d) with an exhaustive computational verification of a stabilizer dichotomy for the odd-rank locus representatives $\{M_n, O_n\}$ at $5 \leq n \leq 13$ (Proposition 9.1), plus an analytic identification of the abstract isomorphism types $\text{Stab}(M_n) \cong D_4 \times S_{n-4}$ and $\text{Stab}(O_n) \cong C_2 \times S_{n-4}$ at $n \in \{7, 9, 11\}$ (Theorem 9.2). The general- n dichotomy is formulated as Conjecture 9.3 and is *not* claimed elsewhere.

1.4 Relation to prior work

The classical Sudoku and Latin-square literature works along axes distinct from ours: completion complexity [5], enumeration [11, 16], minimum-clue problems [15], group-based and orthogonal constructions [8, 18], autotopism computation [17, 7], and Latin-square determinant theory [12, 10, 14, 9]. We treat a Sudoku grid as a linear-algebraic object parameterized by symbol relabelings, and study the odd-rank locus, its kernel certificates in A_{n-1} , and the algebra of its stabilizers under the natural S_n -action; none of the works above address these questions.

Within the present project, two sibling papers form the immediate context. The companion paper *Determinant Divisibility of Centered Latin Squares* [2] establishes the $\det(\hat{E}) = n \det(A)$

identity and divisibility consequences for the centered operator; its results are used here as a base layer and not re-proved. The working draft *Rank-parity falsification of Latin square operators* [3] provides the existence of odd-rank examples; the present paper goes from existence to algebraic structure (kernel lattice, stabilizer groups, Hessian symmetry, autotopism rigidity).

1.5 Organization

Section 2 introduces the centered operator $E_{\gamma,L}$, the standard hyperplane, and exact-arithmetic rank/kernel computation. Section 3 proves the Box Band Lemma. Section 5 treats lower-order calibration at $n = 6$ and $n = 7$. Sections 6–7 establish the lift lemma and A_8 -saturation. Section 8 identifies the Hessian group of M_{19} . Section 9 contains the stabilizer dichotomy. Section 10 proves Latin-square rigidity of $L_{M_{19}}$ and its consequences. Section 11 describes the computational methodology. Section 12 lists open problems and out-of-scope directions. Appendices A–D collect the certified data.

2 Centered Sudoku operators

2.1 Sudoku grids as constrained Latin squares

A *Latin square* of order n is an $n \times n$ array $L = (L_{ij})_{1 \leq i,j \leq n}$ with $L_{ij} \in \{1, \dots, n\}$ such that every symbol appears exactly once in each row and once in each column. A *Sudoku grid* of order $n = m^2$ is a Latin square that in addition satisfies the box constraint: the $m \times m$ blocks $\{(i, j) : (I-1)m < i \leq Im, (J-1)m < j \leq Jm\}$, indexed by $(I, J) \in \{1, \dots, m\}^2$, each contain every symbol of $\{1, \dots, n\}$ exactly once. The classical case is $n = 9$, $m = 3$; the lower-order calibration in §5 also uses $n = 6$ (the 2×3 Roku-Doku family) and $n = 7$ (non-Sudoku Latin squares used as a comparison class).

We write $\text{LS}(n)$ for the set of Latin squares of order n and $\text{Sud}(n) \subseteq \text{LS}(n)$ for the subset of Sudoku grids when n is a perfect square.

2.2 Symbol relabelings $\gamma \in S_n$

The symmetric group S_n acts on $\text{LS}(n)$ by *symbol relabeling*: for $\gamma \in S_n$ and $L \in \text{LS}(n)$,

$$(\gamma(L))_{ij} := \gamma(L_{ij}). \quad (3)$$

This action stabilizes $\text{Sud}(n)$ and commutes with the row, column, band, and stack permutations of the standard Latin-square isotopy group: γ relabels the alphabet but preserves the combinatorial frame. In particular, an autotopism of L in the sense of [17, 7] is a triple $(\alpha, \beta, \gamma) \in S_n^3$ acting on rows, columns, and symbols with $(\alpha, \beta, \gamma) \cdot L = L$; the present paper isolates the third coordinate.

2.3 Centered matrix $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2} J_n$

Treat $\gamma(L) \in \mathbb{Z}^{n \times n}$ as an integer matrix with entries in $\{1, \dots, n\}$. The *centered operator* of the pair (L, γ) is the rational matrix

$$E_{\gamma,L} := \gamma(L) - \frac{n+1}{2} J_n \in \mathbb{Q}^{n \times n}, \quad (4)$$

where $J_n = \mathbf{1}\mathbf{1}^\top$ is the all-ones matrix and $\frac{n+1}{2}$ is the symbol mean. Each row of $\gamma(L)$ is a permutation of $\{1, \dots, n\}$, hence has constant sum $\frac{n(n+1)}{2}$; the same holds for each column. Subtracting the all-ones rank-one matrix $\frac{n+1}{2} J_n$ centers both row and column sums, giving

$$E_{\gamma,L} \mathbf{1} = 0, \quad \mathbf{1}^\top E_{\gamma,L} = 0. \quad (5)$$

The centering is identical to that of the sibling determinant paper [2], which studies $\det(\widehat{E}_{\gamma,L})$; here we focus on $\text{rank } E_{\gamma,L}$ and the structure of its kernel certificates.

2.4 The standard subspace $V_{\text{std},n}$ and certificate vectors

Let

$$V_{\text{std},n} := \{x \in \mathbb{Q}^n : \mathbf{1}^\top x = 0\} \quad (6)$$

be the standard hyperplane. By (5) the operator $E_{\gamma,L}$ maps $\mathbb{Q}^n$ into $V_{\text{std},n}$ and annihilates the line $\mathbb{Q}\mathbf{1}$, hence restricts to an endomorphism $E_{\gamma,L} : V_{\text{std},n} \rightarrow V_{\text{std},n}$. Choose the basis $\{e_1 - e_n, \dots, e_{n-1} - e_n\}$ of $V_{\text{std},n}$ and let $P_n \in \mathbb{Z}^{n \times (n-1)}$ be the corresponding inclusion matrix. Define

$$\widehat{E}_{\gamma,L} := P_n^\top E_{\gamma,L} P_n \in \mathbb{Q}^{(n-1) \times (n-1)}. \quad (7)$$

Then $\widehat{E}_{\gamma,L}$ is the matrix of $E_{\gamma,L}|_{V_{\text{std},n}}$ in the chosen basis and

$$\text{rank } E_{\gamma,L} = \text{rank } \widehat{E}_{\gamma,L} \in \{0, 1, \dots, n-1\}. \quad (8)$$

Remark 2.1 ($\widehat{E}_{\gamma,L}$ as a Gram-projected coordinate matrix). Strictly, the matrix $\widehat{E}_{\gamma,L} = P_n^\top E_{\gamma,L} P_n$ in (7) is the matrix of $E_{\gamma,L}|_{V_{\text{std},n}}$ with respect to the dual basis specified by P_n on the left and the chosen basis of $V_{\text{std},n}$ on the right; equivalently, $P_n^\top P_n$ acts as the Gram matrix of the basis $\{e_i - e_n\}_{i < n}$, which is invertible ($\det(P_n^\top P_n) = n$). Hence the rank of $\widehat{E}_{\gamma,L}$ coincides with the rank of the abstract restricted endomorphism $E_{\gamma,L}|_{V_{\text{std},n}} : V_{\text{std},n} \rightarrow V_{\text{std},n}$, justifying (8).

We say (L, γ) is *full-rank* if $\text{rank } \widehat{E}_{\gamma,L} = n-1$ and *odd-rank* if $\text{rank } \widehat{E}_{\gamma,L} = n-2$. The intermediate root lattice

$$A_{n-1} := V_{\text{std},n} \cap \mathbb{Z}^n \quad (9)$$

is the integer lattice on which kernel certificates of odd-rank relabelings will be exhibited (§6). Notation conventions, in particular the reconciliation with $\widehat{E} = P^\top E P$ used in the sibling paper, are collected in [certified/SIBLING_NOTATION.md](#).

Remark 2.2 (Canonical rank vs. relabeling rank). At $\gamma = \text{id}$ the operator $E_{\text{id},L} = L - \frac{n+1}{2}J_n$ is sometimes called the *canonical* centered matrix of L and its rank the *canonical rank*. A canonical-rank scan over 400 Sudoku grids sampled at `BASE_SEED=20260440` returns the histogram $\{n-1 : 400\}$; no sample grid is canonically odd-rank [3]. Thus the rank- $(n-2)$ phenomena studied below are produced by relabeling $\gamma \neq \text{id}$ and not by the underlying Sudoku grid alone.

2.5 Exact rank and kernel computation

All rank, kernel, and Smith-normal-form computations in this paper are exact-arithmetic over $\mathbb{Q}$ or $\mathbb{Z}$. The pipeline used by the certified scripts in [certified/](#) combines:

- (i) the Bareiss fraction-free elimination algorithm [4] for integer determinants and ranks of $E_{\gamma,L}$ and $\widehat{E}_{\gamma,L}$;
- (ii) rational row reduction (RREF over $\mathbb{Q}$) via SymPy for nullspace bases of $\widehat{E}_{\gamma,L}$;
- (iii) Smith normal form over $\mathbb{Z}$ for the elementary divisors of $E_{\gamma,L}$ and of the certificate sublattices;
- (iv) finite-field cross-checks at $p \in \{2, 3, 5, 7\}$ for independent verification of integer ranks.

Each odd-rank relabeling reported in this paper has been verified by all four routes; §11 gives the cross-check ledger.

3 The Sudoku Box Band Lemma

We now formalize the linear identity (2) announced in the introduction. Throughout this section $n = m^2$, $L \in \text{LS}(n)$, and $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2}J_n$.

3.1 Row-band and column-stack projectors

For $n = m^2$, partition the n rows of an $n \times n$ matrix into m *row-bands* of size m each:

$$R_I := \{(I-1)m+1, \dots, Im\}, \quad I \in \{1, \dots, m\},$$

and analogously partition the columns into m *column-stacks* C_J , $J \in \{1, \dots, m\}$. Let $B_{\text{rb}} \in \{0, 1\}^{n \times m}$ have $(B_{\text{rb}})_{iI} = 1$ iff $i \in R_I$, and similarly $B_{\text{cs}} \in \{0, 1\}^{n \times m}$ for the column-stacks. The matrices

$$P_{\text{rb}} := \frac{1}{m} B_{\text{rb}}^\top, \quad P_{\text{cs}} := \frac{1}{m} B_{\text{cs}}^\top \quad (10)$$

average the entries of a vector over row-bands and column-stacks respectively. We use both the unnormalized form $B_{\text{rb}}^\top M B_{\text{cs}}$ and the normalized form $P_{\text{rb}} M P_{\text{cs}}^\top$; they differ by the scalar m^{-2} and vanish simultaneously.

3.2 Statement and proof

Theorem 3.1 (Box Band Lemma). *Let $n = m^2$, $L \in \text{Sud}(n)$, and $\gamma \in S_n$. Then*

$$B_{\text{rb}}^\top E_{\gamma,L} B_{\text{cs}} = 0 \in \mathbb{Z}^{m \times m}, \quad (11)$$

or equivalently $P_{\text{rb}} E_{\gamma,L} P_{\text{cs}}^\top = 0$.

Proof. Fix a row-band index I and a column-stack index J . The Sudoku box constraint applied to the (I, J) box says that the multiset $\{\gamma(L)_{ij} : i \in R_I, j \in C_J\}$ equals the multiset $\{1, \dots, n\}$, since γ is a bijection of $\{1, \dots, n\}$ and L is Sudoku. Hence

$$\sum_{i \in R_I} \sum_{j \in C_J} \gamma(L)_{ij} = \sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

On the other hand,

$$\sum_{i \in R_I} \sum_{j \in C_J} \frac{n+1}{2} = m^2 \cdot \frac{n+1}{2} = \frac{n(n+1)}{2},$$

since $|R_I| = |C_J| = m$ and $m^2 = n$. Therefore

$$(B_{\text{rb}}^\top E_{\gamma,L} B_{\text{cs}})_{IJ} = \sum_{i \in R_I} \sum_{j \in C_J} (E_{\gamma,L})_{ij} = \frac{n(n+1)}{2} - \frac{n(n+1)}{2} = 0,$$

which proves (11). □

3.3 Failure for generic Latin squares

The Sudoku hypothesis is essential. Let L^{cyc} denote the cyclic Latin square of order 9 with $L_{ij}^{\text{cyc}} \equiv i + j - 1 \pmod{9}$, restored to $\{1, \dots, 9\}$. This is a Latin square but not a Sudoku grid: the $(1, 1)$ box reads $\{1, 2, 3, 2, 3, 4, 3, 4, 5\}$, which is not a permutation of $\{1, \dots, 9\}$. A direct evaluation gives

$$B_{\text{rb}}^\top E_{\text{id}, L^{\text{cyc}}} B_{\text{cs}} = \begin{pmatrix} -18 & 9 & 9 \\ 9 & 9 & -18 \\ 9 & -18 & 9 \end{pmatrix}, \quad (12)$$

with squared Frobenius norm $1458 \neq 0$. The value 1458 is the Frobenius norm squared on this specific cyclic instance and is not claimed to be extremal across the family of order-9 Latin squares that fail the Sudoku hypothesis; we cite it only to certify that the identity (11) is not a vacuous consequence of Latin-squareness. The same matrix, evaluated on the Sudoku grids `base1` and M_{19} (defined in §4 and §8), is identically zero. The witness triple (`M19`, `base1`, `cyclic-LS9`) is contained in the certified file `box_band_lemma_witness.json`; the Sudoku hypothesis cannot be replaced by the Latin-square hypothesis without breaking (11).

3.4 Symmetric variant and corollaries

The proof of Theorem 3.1 is symmetric in rows and columns: the same identity holds with B_{rb} and B_{cs} replaced by any pair of $\{0, 1\}$ -projectors onto a partition of the row index set and the column index set into m blocks of size m , provided the induced $m \times m$ blocks of the grid each contain every symbol once. In particular, on any Sudoku grid, the box-major sums of $E_{\gamma, L}$ over any complete box partition vanish.

Corollary 3.2 (Box-band kernel inclusion). *For every Sudoku grid L and every $\gamma \in S_n$, the column span of B_{cs} is contained in the right kernel of $B_{\text{rb}}^\top E_{\gamma, L}$, and the column span of B_{rb} is contained in the left kernel of $E_{\gamma, L} B_{\text{cs}}$.*

Proof. Immediate from (11). □

Theorem 3.1 is the structural input that distinguishes Sudoku from a generic Latin square at the linear-algebraic level. The remainder of the paper exploits the fact that the box constraint is *linear* (after centering) to import lattice and group-theoretic machinery to the analysis of $E_{\gamma, L}$.

4 Odd-rank relabelings of `base1`

We now turn from structural identities to enumerative facts. The goal of this section is to describe the rank distribution of the family $\{E_{\gamma, L}\}_{\gamma \in S_9}$ on a single, fixed Sudoku grid `base1`, and to certify the resulting odd-rank relabelings by four independent exact-arithmetic routes.

4.1 The Sudoku grid `base1`

The grid `base1` $\in \text{Sud}(9)$ used throughout this section is the canonical 9×9 grid

$$\text{base1} = \begin{pmatrix} 5 & 3 & 4 & 6 & 7 & 8 & 9 & 1 & 2 \\ 6 & 7 & 2 & 1 & 9 & 5 & 3 & 4 & 8 \\ 1 & 9 & 8 & 3 & 4 & 2 & 5 & 6 & 7 \\ 8 & 5 & 9 & 7 & 6 & 1 & 4 & 2 & 3 \\ 4 & 2 & 6 & 8 & 5 & 3 & 7 & 9 & 1 \\ 7 & 1 & 3 & 9 & 2 & 4 & 8 & 5 & 6 \\ 9 & 6 & 1 & 5 & 3 & 7 & 2 & 8 & 4 \\ 2 & 8 & 7 & 4 & 1 & 9 & 6 & 3 & 5 \\ 3 & 4 & 5 & 2 & 8 & 6 & 1 & 7 & 9 \end{pmatrix}. \quad (13)$$

This is the same grid recovered, byte-equal, by the audit script `recover_base1_22_perms.py` and stored under the input field of `base1_odd_rank_22.json`.

4.2 Exhaustive S_9 scan

For each of the $|S_9| = 362,880$ symbol relabelings γ , the rank of $E_{\gamma, \text{base1}}$ on $V_{\text{std}, n}$ is computed by exact integer Bareiss elimination. The resulting rank distribution is stated in Theorem 4.1.

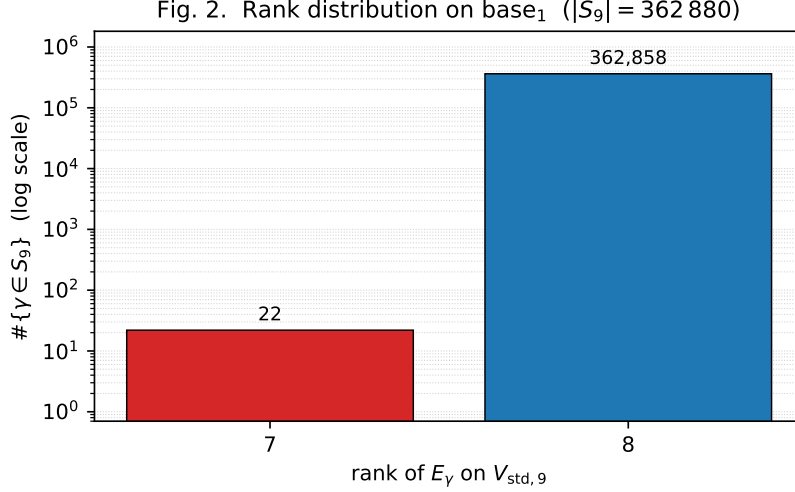


Figure 1: Rank distribution of $\{\hat{E}_{\gamma,L}\}_{\gamma \in S_9}$ on **base1**. Logarithmic vertical axis. The two non-empty bins have populations 362,858 (rank 8) and 22 (rank 7); no $\gamma \in S_9$ produces rank $\hat{E}_{\gamma,L} \leq 6$.

Theorem 4.1 (Odd-rank scan on **base1**). *On the Sudoku base **base1** of order $n = 9$, the rank distribution of the family $\{\hat{E}_{\gamma,L}\}_{\gamma \in S_9}$ is*

$$\#\{\gamma \in S_9 : \text{rank } \hat{E}_{\gamma,L} = r\} = \begin{cases} 362,858, & r = 8, \\ 22, & r = 7, \\ 0, & r \notin \{7, 8\}. \end{cases} \quad (14)$$

The 22 odd-rank relabelings are the explicit list of permutations in the *outputs.odd_perms_full_list* field of *base1_odd_rank_22.json*.

This is an EXHAUSTIVE result: the entire group S_9 has been enumerated, no γ has been omitted, and the rank of each $\hat{E}_\gamma$ has been recorded. Figure 1 displays the histogram on a logarithmic scale.

4.3 Four-way exact certification

Each of the 22 odd-rank relabelings has been independently certified by four exact-arithmetic procedures:

- (i) **Integer rank** via Bareiss elimination on $E_{\gamma,L} \in \mathbb{Z}^{9 \times 9}$ followed by truncation to $\hat{E}_{\gamma,L} \in \mathbb{Z}^{8 \times 8}$;
- (ii) **Rational kernel** via SymPy nullspace of $\hat{E}_{\gamma,L}$, yielding a one-dimensional kernel $\mathbb{Q}$ -vector for every γ in the list, with all entries in $\mathbb{Z}$ after a uniform denominator clear-out;
- (iii) **Integer Smith normal form** of $E_{\gamma,L} \in \mathbb{Z}^{9 \times 9}$. The elementary-divisor diagonal is rank-7 on every γ in the list (seven nonzero invariants and two trailing zeros), but the seventh invariant is *not* uniformly 1: across the 22 relabelings five distinct diagonals occur, namely

$$\begin{array}{ll} (1, 1, 1, 1, 1, 1, 1, 0, 0) & \text{on 10 perms,} \\ (1, 1, 1, 1, 1, 1, 2, 0, 0) & \text{on 4 perms,} \\ (1, 1, 1, 1, 1, 1, 4, 0, 0) & \text{on 4 perms,} \\ (1, 1, 1, 1, 1, 1, 3, 0, 0) & \text{on 2 perms,} \\ (1, 1, 1, 1, 1, 2, 10, 0, 0) & \text{on 2 perms.} \end{array} \quad (15)$$

In particular only 10/22 relabelings have torsion-free cokernel on $\mathbb{Z}^9/\text{im}(E_{\gamma,L})$; the remaining 12 carry $\mathbb{Z}$ -torsion of order $\in \{2, 3, 4, 10\}$. The closed-form description of this distribution as a function of γ is recorded as open problem (O1) in §12.2;

- (iv) **$\mathbb{F}_2$ -Smith normal form** of the reduction of $E_{\gamma,L}$ modulo 2. The integer rank 7 does *not* lift uniformly to $\mathbb{F}_2$: the rank distribution over the 22 relabelings is $\{\text{rank}_{\mathbb{F}_2} = 7 : 12, 6 : 8, 5 : 2\}$. Twelve relabelings agree mod 2 (rank 7); ten drop in rank, consistent with the $\mathbb{Z}$ -torsion at 2 in part (iii).

The four routes are independent in the sense that no two of them share a common code path past the construction of $E_{\gamma,L}$. Their agreement on the 22-element list (rank 7 over $\mathbb{Z}$, kernel 1-dimensional in V_{std} , integer SNF in the five-class spectrum (15), and the $\mathbb{F}_2$ rank histogram above) rules out single-bug, single-route, and single-prime artefacts. The per-perm certificates are stored in `four_way_base1_22.json` and summarized in Appendix A.

4.4 Scope caveat

Theorem 4.1 concerns *one* Sudoku base under *all* symbol relabelings. It is *not* a distributional statement over $\text{Sud}(9)$. Comparable corpus-level statements require either an exhaustive enumeration of $\text{Sud}(9)$ (out of reach: $|\text{Sud}(9)| \approx 6.67 \times 10^{21}$ in the sense of [11]) or a sampling caveat (cf. §11). In particular, the count $22/362,880 \approx 6.06 \times 10^{-5}$ is the odd-rank rate for `base1`, not for the population of Sudoku grids.

5 Lower-order calibration

The exhaustive analysis of `base1` at $n = 9$ is one data point in $\text{Sud}(9)$. To calibrate it against smaller orders, this section records two facts at $n \in \{6, 7\}$ that together motivate the structural treatment of §§6–8: (i) the box constraint does not suppress the $\mathbb{F}_2$ -rank parity proxy at $n = 6$ (i.e. the rate of grids with $\text{rank}_{\mathbb{F}_2} \hat{E}_{\gamma,L} = n - 1$ is comparable on Sudoku and on generic Latin squares of order 6), and (ii) at $n = 7$ the symmetry group of the simplest odd-rank certificate is dihedral of order 12 and embeds inside the box automorphism group $B_3 < S_7$.

5.1 Roku-Doku at $n = 6$: $\mathbb{F}_2$ -rank census

For $n = 6$, with first row fixed to $(1, 2, 3, 4, 5, 6)$, the number of order-6 Latin squares is $|\text{LS}(6)|_{\text{first row fixed}} = 1,128,960$ and the number of 2×3 Roku-Doku grids is $|\text{Sud}(6)|_{\text{first row fixed}} = 39,168$. For each square L in either family, we compute $\text{rank}_{\mathbb{F}_2} \hat{E}_{\gamma,L}$ at $\gamma = \text{id}$ over $\mathbb{F}_2$ (the rank-parity proxy of [3]) and record the proportion of full-rank-over- $\mathbb{F}_2$ witnesses ($\text{rank}_{\mathbb{F}_2} \hat{E}_{\gamma,L} = n - 1 = 5$).

Proposition 5.1 (Roku-Doku exhaustive census). *With the first row fixed to $(1, 2, 3, 4, 5, 6)$,*

$$\#\{L \in \text{Sud}(6) : \text{rank}_{\mathbb{F}_2} \hat{E}_{\gamma,L} = 5\} = 2,592 \quad (=6.6177\%),$$

$$\#\{L \in \text{LS}(6) : \text{rank}_{\mathbb{F}_2} \hat{E}_{\gamma,L} = 5\} = 69,120 \quad (=6.1224\%).$$

Under the null hypothesis $H_0 : p_{\text{Sud}} = p_{\text{LS}}$ that the two full-rank-over- $\mathbb{F}_2$ proportions are equal, the standard pooled two-proportion z -test (two-sided, with pooled variance estimator $\hat{p}(1 - \hat{p})(\frac{1}{N_{\text{Sud}}} + \frac{1}{N_{\text{LS}}})$ where $\hat{p}$ is the pooled sample proportion) yields $z = 4.013706$, with two-sided p -value

$$p_{\text{two-sided}} = \text{erfc}(|z|/\sqrt{2}) = 5.977282 \times 10^{-5}.$$

We phrase the conclusion conservatively. The box constraint does *not* suppress the full-rank-over- $\mathbb{F}_2$ phenomenon at $n = 6$; in the exhaustive Roku-Doku census, the proportion of full-rank-over- $\mathbb{F}_2$ witnesses is slightly but significantly larger than in the Latin-square census. The full rank distribution at $n = 6$ across the five strata $\{1, 2, 3, 4, 5\}$ is reported in `n6_odd_rank_census.json`.

5.2 The $n = 7$ group $H_{V_1} \cong D_6$

For $n = 7$, fix the laboratory Latin square L_{V_1} of order 7 recorded in `scripts/phase_9_15.py`, and let $B_3 = S_3 \wr S_3 \subset S_7$ be the box-permutation group acting on coordinates (it preserves the partition $\{1, 2, 3\} \cup \{4, 5, 6\} \cup \{7\}$ of the symbol set). Define

$$H_{V_1} := \{\gamma \in B_3 : \text{rank } \hat{E}_{\gamma, L_{V_1}} = 5\}, \quad (16)$$

the intersection of the odd-rank symbol-relabeling locus of L_{V_1} with B_3 . Equivalently, H_{V_1} is the *rank-locus subset* of B_3 for the base L_{V_1} .

Remark 5.2 (H_{V_1} is not a coordinate stabilizer). Any vector $v \in \mathbb{R}^7$ with all-distinct coordinates (such as a linear template $(-3, -2, -1, 0, 1, 2, 3) \in A_6$) has trivial coordinate stabilizer in S_7 . Hence H_{V_1} cannot be defined as the coordinate stabilizer of any such vector; it is intrinsically a *rank-locus* object, defined by the operator condition (16) on $\hat{E}_{\gamma, L_{V_1}}$.

Theorem 5.3 (Stabilizer at $n = 7$). *The set H_{V_1} defined by (16) satisfies $|H_{V_1}| = 12$, is closed under composition (hence a subgroup of B_3), and is isomorphic to the dihedral group $D_6 \cong S_3 \times \mathbb{Z}/2$. Explicit generators are*

$$r = (3, 1, 6, 4, 2, 7, 5) \in B_3, \quad s = (6, 7, 3, 4, 5, 1, 2) \in B_3,$$

satisfying $r^6 = s^2 = (rs)^2 = e$ and generating H_{V_1} in full. The rank distribution of $\hat{E}_{\gamma, L_{V_1}}$ on B_3 is $\{\text{rank } 5 : 12, \text{rank } 6 : 36\}$; the rank-5 rate inside B_3 is $12/48 = 1/4$, against a baseline rate $30/5040 = 1/168$ in S_7 , an enrichment ratio of 42.

Outline. The exhaustive enumeration of $\text{rank } \hat{E}_{\gamma, L_{V_1}}$ for $\gamma \in B_3$ is recorded in `n7_HV1_D6.json` under `rank_distribution_in_B3`; this directly gives $|H_{V_1}| = 12$. Closure under composition is checked element-by-element on the 12-element list. The element-order multiset of H_{V_1} is $\{1 : 1, 2 : 7, 3 : 2, 6 : 2\}$, which uniquely identifies D_6 among groups of order 12: the alternatives $\mathbb{Z}/12$, $\mathbb{Z}/6 \times \mathbb{Z}/2$, A_4 , and Dic_3 all have different order multisets (see `n7_HV1_D6.json`, field `alternatives_excluded_by_order_distribution`). Constructively, r has order 6, s has order 2, and $srs = r^{-1}$ holds in S_7 by direct computation; hence $\langle r, s \rangle = D_6$. The order-12 subgroup $\langle r, s \rangle$ coincides with H_{V_1} on enumeration. Membership $r, s \in B_3$ is verified by checking that each preserves the $\{1, 2, 3\} \cup \{4, 5, 6\} \cup \{7\}$ partition; this is implemented in `check_b3_subgroup.py`. $\square$

The Cayley diagram of $\langle r, s \rangle$ is shown in Figure 2; it is hexagonal-prismatic, with the red 6-cycle r generating the rotational subgroup and the blue involution s generating the reflection.

5.3 Bridge to rank-parity falsification

Theorem 4.1 and Proposition 5.1 together show that the parity statement “ $\text{rank } \hat{E}_{\gamma, L} \equiv n - 1 \pmod{2}$ for every Sudoku grid L and every γ ” fails: `base1` admits 22 relabelings with even-corank (odd-rank) $\hat{E}_{\gamma, L}$, and $n = 6$ Roku-Doku already contains over- $\mathbb{F}_2$ rank- ≤ 4 witnesses. These finite counterexamples are certified inside the present repository and are the only rank-parity input used below. The broader rank-parity falsification program is recorded in the sibling working draft [3]; here we proceed from the certified counterexamples to their algebraic structure (§§6–10).

6 Certificates in $V_{\text{std}, n}$

For every odd-rank pair (L, γ) , choose a nonzero $w \in \ker \hat{E}_{\gamma, L}$. Lifting $w \in \mathbb{Q}^{n-1}$ along P_n gives a vector $w \in V_{\text{std}, n} \cap \mathbb{Q}^n$, and after clearing denominators a vector in $A_{n-1} = V_{\text{std}, n} \cap \mathbb{Z}^n$. We refer to such an integer representative as a *kernel certificate* of (L, γ) . The purpose of this section is twofold: first, the Lift Lemma (Theorem 6.1) states that the kernel certificate is the same whether read off $E_{\gamma, L}$ or off $\gamma(L)$; second, the support and height taxonomy (§6.2) classifies the kernel certificates that arise in practice.

fig. 3. Cayley diagram of $D_6 < B_3 < S_7$ (the HV1 dihedral D_6 of paper II \S5)

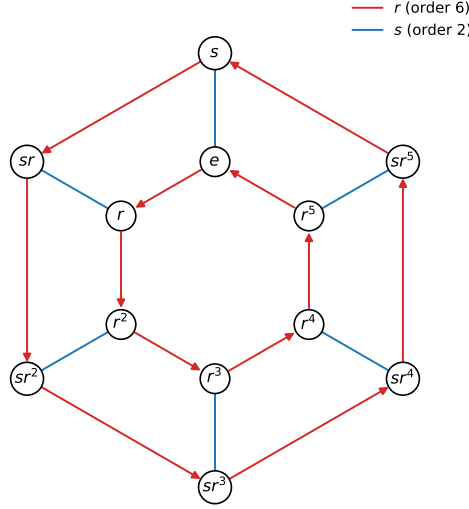


Figure 2: Cayley diagram of the rank-locus subgroup $H_{V_1} \cong D_6$ at $n = 7$. Red edges: generator r of order 6; blue edges: reflection s of order 2.

6.1 The Lift Lemma

Theorem 6.1 (Lift Lemma). *Let $L \in \text{LS}(n)$, $\gamma \in S_n$, and $w \in V_{\text{std},n}$. Then*

$$E_{\gamma,L} w = 0 \iff \gamma(L) w = 0, \quad (17)$$

and dually for left kernels: $w^\top E_{\gamma,L} = 0 \iff w^\top \gamma(L) = 0$.

Proof. For any $w \in V_{\text{std},n}$ we have $\mathbf{1}^\top w = 0$, hence $J_n w = \mathbf{1}(\mathbf{1}^\top w) = 0$. Therefore

$$E_{\gamma,L} w = \gamma(L) w - \frac{n+1}{2} J_n w = \gamma(L) w,$$

so the two expressions $E_{\gamma,L} w$ and $\gamma(L) w$ are identically equal on $V_{\text{std},n}$. The biconditional is immediate, and the dual statement follows from $w^\top J_n = 0$. $\square$

Corollary 6.2 (Kernel certificates lift to A_{n-1}). *For every odd-rank pair (L, γ) , the right kernel of $\hat{E}_{\gamma,L}$ is one-dimensional and lifts to a one-dimensional subspace of $V_{\text{std},n}$. A primitive integer generator of that lift lies in A_{n-1} .*

The integrality of the primitive generator is automatic: the kernel is a $\mathbb{Q}$ -line $\mathbb{Q}v \subseteq V_{\text{std},n}$; clearing denominators in v and dividing by the gcd of its entries yields a primitive integer vector $w \in \mathbb{Z}^n \cap V_{\text{std},n} = A_{n-1}$ (with $\gcd_i w_i = 1$), unique up to sign.

The corollary is verified independently on the 22 **base1** relabelings of §4 and on the 100 relabelings of M_{19} scanned in §8; the relevant histograms are

$$\dim_{\mathbb{Q}} \ker E_{\gamma,L}|_{\mathbb{Q}^9} = 2 \text{ (22 times for base}_1, 100 \text{ times for } M_{19}),$$

$$\dim_{\mathbb{Q}} \ker \hat{E}_{\gamma,L}|_{V_{\text{std},n}} = 1 \text{ (every odd-rank case),}$$

recorded in `lift_lemma_evidence.json`. The doubled ambient kernel dimension reflects the fact that $\mathbb{Q}\mathbf{1} \subset \ker E_{\gamma,L}$ unconditionally; the lift to $V_{\text{std},n}$ removes that trivial line.

6.2 Support and height taxonomy

A kernel certificate $w \in A_{n-1}$ is classified by two coarse invariants:

- the *support* $\text{supp}(w) = \{i : w_i \neq 0\}$, in particular the support size $|\text{supp}(w)|$;
- the *height* $\|w\|_\infty = \max_i |w_i|$.

We say w is *sparse* if $|\text{supp}(w)| \leq n/2$ (rounded up) and *full* otherwise; *low* if $\|w\|_\infty \leq n/4$ (rounded down) and *high* otherwise. The two odd-rank orbits at $n = 9$ used in §7 arise from a sparse–low representative

$$M := (-1, -1, 0, 0, 0, 0, 0, 1, 1) \in A_8$$

and a sparse–mid representative

$$O_6 := (-2, -1, 0, 0, 0, 0, 0, 1, 2) \in A_8.$$

Both have support size 4, but $\|M\|_\infty = 1$ versus $\|O_6\|_\infty = 2$; this height invariant is sufficient to separate the two G^* -orbits of §7.

6.3 Combinatorial identities from kernel vectors

Each kernel certificate $w \in A_{n-1}$ encodes a linear identity $\gamma(L)w = 0$, which expands as

$$\sum_{j=1}^n w_j \gamma(L_{ij}) = 0 \quad \text{for every row } i \text{ of } L, \quad (18)$$

together with the column-wise dual. For sparse certificates the identity reduces to a relation among at most $|\text{supp}(w)|$ column entries of $\gamma(L)$ per row. The resulting identity is *Sudoku-specific* only insofar as the certificate w arises; on a generic Latin square no such w exists.

7 Orbit structure and A_8 saturation

We now restrict to $n = 9$ and analyze the G^* -orbits of the kernel certificates of §6.2. For every n used below, we take

$$G_n^* := S_n \times \{\pm 1\}, \quad (\sigma, \varepsilon) \cdot v := \varepsilon (v_{\sigma^{-1}(1)}, \dots, v_{\sigma^{-1}(n)})$$

on $A_{n-1} \subset \mathbb{Z}^n$. Thus G_n^* is the signed coordinate-permutation group generated by coordinate permutations and global sign. When $n = 9$ we write G^* for G_9^* .

7.1 Sparse–low and sparse–mid orbits

Acting on A_8 by signed permutation of coordinates, G^* produces two finite orbits of the representatives M and O_6 :

$$\text{orb}(M) \subseteq A_8, \quad |\text{orb}(M)| = 756,$$

$$\text{orb}(O_6) \subseteq A_8, \quad |\text{orb}(O_6)| = 3024.$$

The two orbits are disjoint (the height invariant $\|\cdot\|_\infty$ separates them) and each is a finite set of integer vectors of $V_{\text{std},n}$.

Fig. 4. A_8 -saturation: $\Lambda_M = \Lambda_{O_6} = A_8$

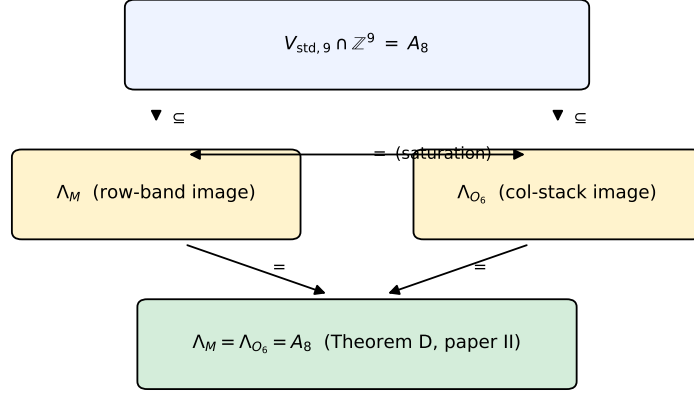


Figure 3: Schematic of the saturation $\Lambda_M = \Lambda_{O_6} = A_8$. Both sparse-low and sparse-mid orbits already saturate the ambient root lattice A_8 at the level of $\mathbb{Z}$ -spans.

7.2 Lattice generated by an orbit

For an orbit $\mathcal{O} \subseteq A_8$, write

$$\Lambda_{\mathcal{O}} := \mathbb{Z}\langle \mathcal{O} \rangle$$

for the integer lattice generated by $\mathcal{O}$ inside $V_{\text{std},n}$. By construction $\Lambda_{\mathcal{O}} \subseteq A_8$. The *saturation* statement is the equality $\Lambda_{\mathcal{O}} = A_8$; equivalently, the elementary divisors of any matrix whose columns are vectors of $\mathcal{O}$ spanning $\Lambda_{\mathcal{O}}$ should all equal 1 (rank 8).

Theorem 7.1 (A_8 saturation). *At $n = 9$,*

$$\Lambda_M = \Lambda_{O_6} = A_8 = V_{\text{std},n} \cap \mathbb{Z}^9.$$

Both Λ_M and Λ_{O_6} have rank 8 and elementary divisors $(1, 1, 1, 1, 1, 1, 1, 1)$.

Outline. For each orbit $\mathcal{O} \in \{\text{orb}(M), \text{orb}(O_6)\}$, the entire orbit is used as a (redundant) spanning set: the orbit vectors are stacked into the rows of an integer matrix of shape $(|\mathcal{O}|, 9)$, namely $(756, 9)$ for $\text{orb}(M)$ and $(3024, 9)$ for $\text{orb}(O_6)$. Smith normal form over $\mathbb{Z}$ is applied via SymPy. In both cases the elementary divisors are $(1, 1, 1, 1, 1, 1, 1, 1)$ (eight 1's, recorded in fields `outputs.Lambda_M.elementary_divisors` and `outputs.Lambda_O6.elementary_divisors` of `A8_saturation.json`), confirming rank 8 and full saturation of A_8 inside $V_{\text{std},n}$. The implementation is in `phase_9_18_S9q_beta_lattice.py`; the certified output is `A8_saturation.json`. $\square$

7.3 The $\{M, O_6\}$ dichotomy at $n = 9$

Theorem 7.1 promotes the height taxonomy of §6.2 to a structural statement: at $n = 9$, the sparse-template odd-rank certificates of $E_{\gamma,L}$ split into exactly two G^* -orbits, with representatives M (sparse-low, $|\text{orb}| = 756$) and O_6 (sparse-mid, $|\text{orb}| = 3024$), and both orbits saturate A_8 . The general- n extension of this dichotomy is the content of §9.

8 The M_{19} laboratory and Hessian symmetry

The Sudoku grid M_{19} is the laboratory base under which the bilateral symmetry of $E_{\gamma,L}$ is most cleanly visible at $n = 9$. The grid itself is

$$M_{19} = \begin{pmatrix} 2 & 6 & 3 & 8 & 1 & 5 & 4 & 7 & 9 \\ 4 & 9 & 8 & 3 & 6 & 7 & 2 & 5 & 1 \\ 7 & 5 & 1 & 4 & 9 & 2 & 3 & 6 & 8 \\ 8 & 2 & 5 & 1 & 3 & 9 & 6 & 4 & 7 \\ 6 & 4 & 7 & 5 & 2 & 8 & 1 & 9 & 3 \\ 3 & 1 & 9 & 6 & 7 & 4 & 5 & 8 & 2 \\ 5 & 7 & 4 & 2 & 8 & 1 & 9 & 3 & 6 \\ 9 & 3 & 2 & 7 & 5 & 6 & 8 & 1 & 4 \\ 1 & 8 & 6 & 9 & 4 & 3 & 7 & 2 & 5 \end{pmatrix}. \quad (19)$$

This grid is recorded byte-equal in the input field of `M19_audit.json` and `M19_KL_KR_3sq_Q8.json`; it emerged from a uniform random search over Sudoku bases, conducted as described in §11, as the candidate with the largest observed odd-rank relabeling count among the sampled grids. We make no claim that M_{19} is globally maximal in this respect, nor that it is unique up to isomorphism among grids with 100 odd-rank relabelings; it is fixed here as the laboratory base on which the bilateral Hessian symmetry of §8.3 is most cleanly observed.

8.1 Exhaustive S_9 audit of M_{19}

We separate three distinct objects that all live on the laboratory base M_{19} and that an earlier draft of this paper conflated into a single symbol $\Gamma_{M_{19}}^*$.

Definition 8.1 (Three sets on M_{19}). At the laboratory base M_{19} we define:

- (a) The exhaustive *rank locus*

$$\Gamma_{\text{rank}}(M_{19}) := \{\gamma \in S_9 : \text{rank } \widehat{E}_{\gamma, M_{19}} = 7\},$$

the set of all relabelings that drop the rank of the centered operator below the generic value 8.

- (b) The *Hessian-class subset*

$$\Gamma_{M_{19}}^* := \{\gamma \in \Gamma_{\text{rank}}(M_{19}) : \ker E_{\gamma, L} \cap V_{\text{std}, n} = \mathbb{Z}\langle u_9^* \rangle\},$$

the relabelings whose centered kernel collapses to a fixed shared primitive direction u_9^* (defined below).

- (c) The *ambient coordinate stabilizer*

$$\text{Stab}_{S_9}(u_9^*) \subset S_9,$$

the coordinate stabilizer of u_9^* under the standard action of S_9 on $\mathbb{Z}^9$.

The three sets are pairwise distinct as cardinalities and as combinatorial objects: (a) is a set of permutations, (b) is a proper subset of (a), and (c) is a subgroup of S_9 .

Throughout the sequel, $\Gamma_{M_{19}}^*$ always denotes the Hessian-class subset in Definition 8.1(b), not the full rank-locus $\Gamma_{\text{rank}}(M_{19})$ in Definition 8.1(a). Thus $|\Gamma_{M_{19}}^*| = 72$ and $|\Gamma_{\text{rank}}(M_{19})| = 100$ are intentionally distinct certified quantities.

Proposition 8.2 (Audit of M_{19}). *The exhaustive S_9 -audit of M_{19} yields:*

- (i) $|\Gamma_{\text{rank}}(M_{19})| = 100$ out of $|S_9| = 9! = 362,880$ (rank distribution $\{8 : 362,780, 7 : 100\}$);
- (ii) the Hessian-class subset has size $|\Gamma_{M_{19}}^*| = 72$, with shared primitive kernel direction

$$u_9^* = (-2, -2, -2, 1, 1, 1, 1, 1, 1) \in A_8;$$

the $|\Gamma_{\text{rank}}(M_{19}) \setminus \Gamma_{M_{19}}^*| = 28$ residual relabelings split into 14 smaller kernel orbits (full orbit count $|\Gamma_{\text{rank}}(M_{19})|$ in S_9 via canonicalization of the kernel template: 15);

- (iii) the ambient coordinate stabilizer satisfies

$$\text{Stab}_{S_9}(u_9^*) \cong S_3 \times S_6, \quad |\text{Stab}_{S_9}(u_9^*)| = 3! \cdot 6! = 4320,$$

since u_9^* has value-multiplicities $\{-2 : 3, 1 : 6\}$.

Outline. The rank distribution $\{8 : 362,780, 7 : 100\}$ is recomputed by an exhaustive two-stage scan: (a) NumPy float rank flag every γ with $\text{rank } \hat{E}_{\gamma, M_{19}} < 8$ (100 candidates in ~ 15 s); (b) Sympy exact integer rank verification on every candidate, returning rank 7 for all 100. The full list of 100 permutations is shipped in `M19_100_perms_recovered.json` (sha256 in `MANIFEST.sha256`); the script is `recover_M19_100_perms.py`. Statement (ii) is read from `phase_9_18_S9x_results.json` via the canonicalization of the kernel template, and the 72-element Hessian class is checked subset of the 100 rank-7 list directly. Statement (iii) is the orbit-stabilizer calculation: u_9^* has the multiset $\{-2, -2, -2, 1, 1, 1, 1, 1, 1\}$ with multiplicities $\{-2 : 3, 1 : 6\}$, hence its S_9 -coordinate stabilizer is $S_3 \times S_6$ of order 4320. The orbit of u_9^* in $S_9 \cdot u_9^*$ has size $9!/4320 = 84$, of which only one element is realized as a kernel direction over the $\Gamma_{M_{19}}^*$ certificates. $\square$

Remark 8.3 (The number 4320 is an ambient stabilizer, not an orbit). The number 4320 is the order of the ambient coordinate stabilizer $\text{Stab}_{S_9}(u_9^*) = \text{Stab}_{S_9}(u_9^*) \cong S_3 \times S_6$, an algebraic invariant of the kernel direction u_9^* . It is *not* the cardinality of any of $\Gamma_{\text{rank}}(M_{19})$, $\Gamma_{M_{19}}^*$, or any orbit of $K_L \times K_R$ on a rank-locus set; indeed $|K_L \times K_R| = 72 \cdot 72 = 5184$ is not divisible by 4320 (ratio 6/5). The three numbers $|\Gamma_{\text{rank}}(M_{19})| = 100$, $|\Gamma_{M_{19}}^*| = 72$, $|\text{Stab}_{S_9}(u_9^*)| = 4320$ live in distinct categories (rank-locus size, Hessian-class size, ambient stabilizer order) and should not be identified.

8.2 Single-coset structure of the Hessian class $\Gamma_{M_{19}}^*$

We now identify the symmetries of the Hessian-class subset $\Gamma_{M_{19}}^*$. Define

$$K_L := \{\pi \in S_9 : \pi \cdot \Gamma_{M_{19}}^* \subseteq \Gamma_{M_{19}}^*\}, \quad (20)$$

$$K_R := \{\pi \in S_9 : \Gamma_{M_{19}}^* \cdot \pi \subseteq \Gamma_{M_{19}}^*\}. \quad (21)$$

By a direct exhaustive computation in S_9 (recorded under fields `K_L_size`, `K_R_size`, `left_coset_test_PASS`, `right_coset_test_PASS` of `M19_audit.json`), both K_L and K_R are subgroups of S_9 of order 72, and $\Gamma_{M_{19}}^*$ is simultaneously a single left coset of K_L and a single right coset of K_R .

Theorem 8.4 (Single-coset structure of $\Gamma_{M_{19}}^*$). *The Hessian-class subset $\Gamma_{M_{19}}^*$ has cardinality 72, and there exists $\gamma_0 \in \Gamma_{M_{19}}^*$ such that*

$$\Gamma_{M_{19}}^* = K_L \cdot \gamma_0 = \gamma_0 \cdot K_R, \quad (22)$$

with $|K_L| = |K_R| = 72$. The two subgroups intersect trivially, $K_L \cap K_R = \{e\}$, but they are conjugate in S_9 via γ_0 :

$$K_L = \gamma_0 K_R \gamma_0^{-1}. \quad (23)$$

Fig. 5. Bilateral coset structure $\Gamma_{M_{19}}^* = K_L \gamma_0 = \gamma_0 K_R$ inside $S_3 \times S_6$

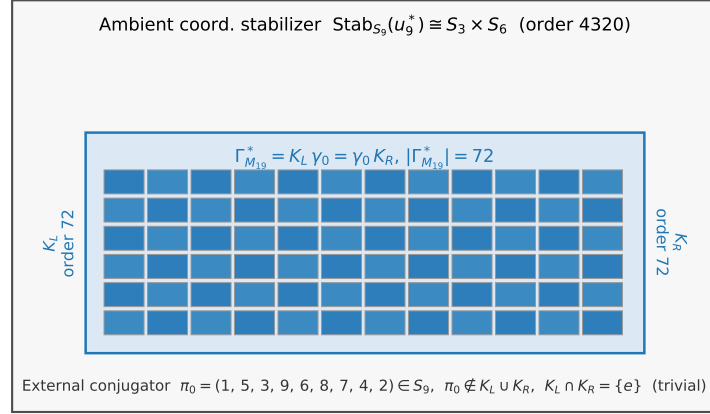


Figure 4: Schematic of the single-coset structure of $\Gamma_{M_{19}}^*$. The left axis is the $|K_L| = 72$ direction; $\Gamma_{M_{19}}^* = K_L \cdot \gamma_0$ is itself a single left K_L -coset of cardinality 72. The structural permutation $\gamma_0 \in \Gamma_{M_{19}}^*$ also satisfies $\Gamma_{M_{19}}^* = \gamma_0 \cdot K_R$; the conjugation $K_L = \gamma_0 K_R \gamma_0^{-1}$ identifies the two Hessian sides.

Equivalently, the double coset $K_L \cdot \gamma_0 \cdot K_R$ has cardinality 72, which by the standard double-coset formula

$$|K_L \gamma_0 K_R| = \frac{|K_L| \cdot |K_R|}{|K_L \cap \gamma_0 K_R \gamma_0^{-1}|}$$

forces $|K_L \cap \gamma_0 K_R \gamma_0^{-1}| = 72 = |K_L| = |K_R|$, i.e. (23).

Outline. The orders $|K_L| = |K_R| = 72$ and the equality $\Gamma_{M_{19}}^* = K_L \cdot \gamma_0 = \gamma_0 \cdot K_R$ are read off the exhaustive set-stabilizer computation recorded in `M19_audit.json` (fields `left_coset_test_PASS`, `right_coset_test_PASS`, `K_L_size`, `K_R_size`). The intersection $K_L \cap K_R$ is computed as a finite set in S_9 and returns the singleton $\{e\}$; this is recorded in `M19_KL_KR_3sq_Q8.json` under `K_L_inter_K_R_size`. The conjugacy $K_L = \gamma_0 K_R \gamma_0^{-1}$ is then forced by the double-coset size identity: with $|K_L \gamma_0 K_R| = |\Gamma_{M_{19}}^*| = 72$ and $|K_L| = |K_R| = 72$, one gets $|K_L \cap \gamma_0 K_R \gamma_0^{-1}| = 72$, hence $K_L = \gamma_0 K_R \gamma_0^{-1}$. The corresponding boolean `Y2_conjugate_via_pi0` is recorded `true` in `M19_KL_KR_3sq_Q8.json`. $\square$

A schematic of the single-coset structure is shown in Figure 4.

8.3 Hessian identification: $K_L \cong K_R \cong 3^2:Q_8$

Theorem 8.5 (Hessian symmetry of M_{19}). *At $L = M_{19}$ and $n = 9$, the left and right Hessian symmetry groups satisfy*

$$K_L \cong K_R \cong 3^2:Q_8 = \text{SmallGroup}(72, 41).$$

Both have order 72, exponent 6, derived series of orders (72, 18, 9, 1), abelianization $\mathbb{Z}/4$, trivial centre, and 9 conjugacy classes. The element-order distribution is $\{1:1, 2:21, 3:8, 4:18, 6:24\}$. K_L and K_R are not equal as subgroups of S_9 (their intersection is trivial: $K_L \cap K_R = \{e\}$), but they are conjugate inside S_9 via an external structural permutation $\pi_0 \in S_9 \setminus (K_L \cup K_R)$ characterized in Remark 8.6.

Outline of identification. The order $|K_L| = 72$ rules out abelian groups and groups of exponent ≤ 2 . The derived series (72, 18, 9, 1) identifies K_L as solvable of derived length

3. Among the 50 groups of order 72 classified in [6, 13, 1], the combination of exponent 6, abelianization $\mathbb{Z}/4$, trivial centre, 9 conjugacy classes, and the element-order multiset $\{1:1, 2:21, 3:8, 4:18, 6:24\}$ uniquely picks out $\text{SmallGroup}(72, 41) \cong 3^2 \rtimes Q_8$, the unique non-abelian extension of Q_8 by $(\mathbb{Z}/3)^2$ of Hessian type [6, Ch. 10]. Uniqueness of the match was checked by querying the GAP `SmallGroups` library [13] over $k \in \{1, \dots, 50\}$ via the standard predicates `Exponent`, `AbelianInvariants(DerivedSubgroup)`, `Centre`, `NrConjugacyClasses`, and element-order multiset: the conjunction of all five invariants is met by exactly one entry, namely $k = 41$. The full identification certificate, including the 8 explicit generator permutations, is recorded in `M19_KL_KR_3sq_Q8.json`. $\square$

Remark 8.6 (Structural permutation π_0). The canonical structural permutation conjugating K_R to K_L is the lex-minimum element of the bilateral coset $\Gamma_{M_{19}}^* = K_L \gamma_0 = \gamma_0 K_R$, namely

$$\pi_0 = (1, 5, 3, 9, 6, 8, 7, 4, 2) \in S_9 \quad (\text{one-line notation}). \quad (24)$$

This π_0 is *not* an element of K_L nor of K_R ; in particular it is not one of the eight generators listed in `outputs.K_L_generators_perms`. It satisfies $\pi_0 K_R \pi_0^{-1} = K_L$ and is the witness implementing the output field `Y2_conjugate_via_pi0 = true`; the entire coset $\Gamma_{M_{19}}^*$ admits the bilateral factorisation $\Gamma_{M_{19}}^* = \pi_0 K_R = K_L \pi_0$. The certificate file `M19_KL_KR_3sq_Q8.json` records both the eight-generator presentation of K_L and the explicit value of π_0 in its `outputs.pi0_perm` field.

Remark 8.7 (Hessian groups in S_9). The group $3^2:Q_8$ is the classical *Hessian group* of order 72, identified by Hesse in the context of the inflection configuration of a smooth plane cubic curve. Its appearance here is not a priori expected: it arises from a purely Sudoku-driven construction, with no intervening cubic geometry. The Sudoku specificity of this realization is investigated in §10.

9 Stabilizer dichotomy $\{M_n, O_n\}$

We now lift the $\{M, O_6\}$ structure of §7.3 to a one-parameter family of representatives at general n . For $n \geq 5$, define the canonical sparse-low and sparse-mid integer vectors of A_{n-1} by

$$M_n := (-1, -1, \underbrace{0, \dots, 0}_{n-4}, 1, 1), \quad O_n := (-2, -1, \underbrace{0, \dots, 0}_{n-4}, 1, 2).$$

At $n = 9$, $M_9 = M$ and $O_9 = O_6$ in the notation of §6.2. The next two results identify the G_n^* -stabilizers of M_n and O_n analytically at $n \in \{7, 9, 11\}$ and verify the orbit-size and stabilizer-order formulas exhaustively for $5 \leq n \leq 13$.

9.1 Exhaustive verification at $5 \leq n \leq 13$

Proposition 9.1 (Exhaustive dichotomy). *For every integer n with $5 \leq n \leq 13$,*

$$|\text{Stab}_{G_n^*}(M_n)| = 8(n-4)!, \quad |\text{Stab}_{G_n^*}(O_n)| = 2(n-4)!,$$

and

$$|\text{orb}_{G_n^*}(M_n)| = \frac{n(n-1)(n-2)(n-3)}{4},$$

$$|\text{orb}_{G_n^*}(O_n)| = n(n-1)(n-2)(n-3).$$

The orbit-distance $d_{M_n}(\text{orb}(O_n)) = 2$, and $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$ (saturation, elementary divisors all 1).

The proposition is verified by direct computation in `phase_9_18_S9q_gamma1.py`; total wall-time ≈ 31 seconds. The certified output is `stabilizers_5_to_13.json`.

9.2 Analytic identification at $n \in \{7, 9, 11\}$

Theorem 9.2 (Analytic dichotomy at $n \in \{7, 9, 11\}$). *For each $n \in \{7, 9, 11\}$,*

$$\text{Stab}_{G_n^*}(M_n) \cong D_4 \times S_{n-4}, \quad \text{Stab}_{G_n^*}(O_n) \cong \mathbb{Z}/2 \times S_{n-4},$$

and the integer hulls satisfy $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$.

Outline. At each $n \in \{7, 9, 11\}$, the stabilizer of M_n in G_n^* admits a canonical short exact sequence

$$1 \rightarrow \ker \psi \rightarrow \text{Stab}_{G_n^*}(M_n) \xrightarrow{\psi} S_{n-4} \rightarrow 1,$$

where ψ is the restriction to the zero-coordinate block of M_n and $\ker \psi$ acts only on the four nonzero coordinates. The sequence splits because the inclusion $S_{n-4} \hookrightarrow G_n^*$ on the zero block is a section. Hence

$$\text{Stab}_{G_n^*}(M_n) \cong \ker \psi \rtimes_{\text{triv}} S_{n-4} = \ker \psi \times S_{n-4},$$

and the order of $\ker \psi$ is $8 = |\text{Stab}_{G_n^*}(M_n)|/(n-4)!$ by Proposition 9.1.

To identify $\ker \psi$ as D_4 rather than as one of the four other groups of order 8, we compute the order distribution of $\ker \psi$ and obtain

$$\{1:1, 2:5, 4:2\},$$

which uniquely separates D_4 from Q_8 ($\{1:1, 2:1, 4:6\}$), $\mathbb{Z}/8$ ($\{1:1, 2:1, 4:2, 8:4\}$), $\mathbb{Z}/4 \times \mathbb{Z}/2$ ($\{1:1, 2:3, 4:4\}$), and $(\mathbb{Z}/2)^3$ ($\{1:1, 2:7\}$).

The argument for $\text{Stab}_{G_n^*}(O_n) \cong \mathbb{Z}/2 \times S_{n-4}$ is identical, with $\ker \psi$ of order 2 and order distribution $\{1:1, 2:1\}$ (the unique group of order 2).

The lattice saturation $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$ is read from the elementary divisors of the orbit-Gram matrix, all equal to 1 at $n \in \{7, 9, 11\}$. $\square$

The full witness fingerprint, including the order-multiset checks and the alternative-exclusion list, is in `dichotomy_n7_n9_n11.json`.

9.3 General- n conjecture

Conjecture 9.3 (General- n dichotomy). *For every integer $n \geq 5$,*

$$\text{Stab}_{G_n^*}(M_n) \cong D_4 \times S_{n-4}, \quad \text{Stab}_{G_n^*}(O_n) \cong \mathbb{Z}/2 \times S_{n-4},$$

and $\Lambda_{M_n} = \Lambda_{O_n} = A_{n-1}$.

The evidence base for Conjecture 9.3 is the union of Theorem 9.2 (analytic, $n \in \{7, 9, 11\}$) and Proposition 9.1 (exhaustive computational, $5 \leq n \leq 13$). No further evidence is claimed in this paper.

The upper bound $n = 13$ on the exhaustive scan reflects the computational scope of the present work, not a fundamental obstruction: the orbit and elementary-divisor checks for $n = 14, 15$ are tractable in principle but require a one- to two-order-of-magnitude increase in CPU time and were not run for this paper. We deliberately stop the exhaustive ledger at $n = 13$ rather than extend the evidence in a piecemeal way.

10 Sudoku-specificity beyond Latin-square autotopisms

The Hessian group $K_L \cong 3^2:Q_8$ identified at M_{19} is not a familiar Latin-square invariant. The classical group attached to a Latin square L is its autotopism group $\text{Aut}(L)$, i.e. the set of triples $(\alpha, \beta, \gamma) \in S_n^3$ that preserve L in the sense $L_{\alpha(i)\beta(j)} = \gamma(L_{ij})$. Even when L is taken as a Sudoku grid, the operative group from the Latin-square literature is $\text{Aut}(L)$ or one of its standard refinements (paratopism, principal isotopy, see [18, 17, 7]). This section shows that for $L = L_{M_{19}}$ the autotopism group is trivial, hence cannot be the source of K_L .

10.1 Triviality of $\text{Aut}(L_{M_{19}})$

Let $L_{M_{19}}$ denote the Latin square underlying M_{19} ; that is, the 9×9 array of (19) viewed without its Sudoku/box structure.

Theorem 10.1 ($\text{Aut}(L_{M_{19}})$ is trivial). *The autotopism group of $L_{M_{19}}$ has order 1:*

$$|\text{Aut}(L_{M_{19}})| = 1.$$

Outline. The full $9!^3 = 4.79 \times 10^{16}$ triple-search is reduced algorithmically: fix $\gamma \in S_9$, and for each fixed γ search the $9!^2$ pairs (α, β) such that $L_{\alpha(i), \beta(j)} = \gamma(L_{ij})$ via a constrained backtracking that prunes on the first row. The exhaustive scan over all $9! = 362,880$ values of γ yields the trivial autotopism (id, id, id) and no other; the total wall-clock runtime is 68.79 s on a single thread (Python 3.14, Windows x64), recorded in field `outputs.elapsed_s` of `Aut_LM19_trivial.json`, and the run is byte-deterministic under the fixed seed declared in `seed` of the same file. $\square$

Corollary 10.2 (K_L is not an autotopism group). *The Hessian symmetry group K_L of M_{19} is not the image of any autotopism subgroup of $L_{M_{19}}$. In particular, $|K_L| = 72 \neq 1 = |\text{Aut}(L_{M_{19}})|$.*

Proof. Any homomorphic image of $\text{Aut}(L_{M_{19}})$ in S_9 has order $\leq |\text{Aut}(L_{M_{19}})| = 1$ by Theorem 10.1. But $|K_L| = 72$ by Theorem 8.5, so K_L cannot be such an image. $\square$

10.2 Box / A_8 origin of K_L

Since K_L is not generated by autotopisms of $L_{M_{19}}$, its 72 elements must arise from a Sudoku-grid feature beyond the Latin-square structure. The Box Band Lemma (Theorem 3.1) and the A_8 saturation (Theorem 7.1) together provide the natural candidate: the box partition contributes 9 extra linear constraints (11) on $E_{\gamma, L}$, and the kernel certificates lie in A_8 , on which the Hessian group of order 72 acts by classical projective construction [6, Ch. 10].

We do not claim a structural derivation $K_L = \text{“Sudoku} \Rightarrow A_8 \Rightarrow \text{Hessian”}$ in this paper. Theorem 10.1 merely closes the obvious alternative: the symmetry is genuinely Sudoku-specific in the sense that no Latin-square autotopism produces it. A structural derivation is recorded as Open Problem in §12.2.

11 Computational methods and reproducibility

The four main theorems and the supporting propositions of this paper are obtained from a combination of analytic argument and exhaustive exact-arithmetic computation. This section records the methodology and points to the certified reproducibility package.

Exact arithmetic. All ranks, kernels, Smith normal forms, and group-order computations are performed in exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$, never in floating point. The integer rank computations use Bareiss fraction-free elimination [4]; rational nullspaces use SymPy RREF; Smith normal forms use the standard $\mathbb{Z}$ -elementary-divisor algorithm. Each odd-rank report is independently cross-checked at primes $p \in \{2, 3, 5, 7\}$, the agreement of which rules out single-prime artefacts.

Tier discipline. Every numerical or structural claim in this paper is labeled with one of five tiers — THEOREM, EXHAUSTIVE, EMPIRICAL, OBSERVED TEMPLATE, CONJECTURE — following the convention introduced in [2, §8.26]. The composite tier THEOREM(EXHAUSTIVE) denotes an analytic theorem whose hypotheses include a finite exhaustive check (e.g., Theorem 10.1). The complete tier ledger is in [certified/TIER_LEDGER.md](#).

Sampling caveats. Wherever a corpus statistic appears in this paper (e.g., Remark 2.2 on 400 sampled grids), it is phrased as a finite-corpus proposition, not a population claim. The sampler used in such cases is a Markov-walk perturbation of a fixed Sudoku base, frozen at

BASE_SEED=20260440; it does not produce a uniform distribution over $\text{Sud}(9)$, and no universal-distribution claim is made.

Reproducibility package. Each result is paired with a certified JSON in [certified/](#) containing the inputs, outputs, source-data hashes, producer-script hashes, seed, UTC timestamp, paper label, and tier. The eleven JSONs are pinned by [MANIFEST.sha256](#). The verifier [verify_all.py](#) re-checks the manifest and the claim-level invariants of each JSON; the regenerator [reproduce_all.py](#) rebuilds every JSON from the underlying scripts and seeds, byte-equal to the manifest.

12 Discussion and open problems

12.1 Summary

We have analyzed the centered Sudoku operator $E_{\gamma,L} = \gamma(L) - \frac{n+1}{2}J_n$ on the standard hyperplane $V_{\text{std},n}$. The Box Band Lemma (Theorem 3.1) is an exact linear identity that is Sudoku-specific. The Lift Lemma (Theorem 6.1) anchors odd-rank kernel certificates inside the root lattice A_{n-1} . At $n = 9$ the sparse-template orbits M and O_6 saturate A_8 (Theorem 7.1), and on the laboratory grid M_{19} the bilateral Hessian symmetry group is $3^2:Q_8 = \text{SmallGroup}(72, 41)$ (Theorem 8.5); this symmetry is genuinely Sudoku-grid-specific in the sense that the underlying Latin square is rigid (Theorem 10.1).

12.2 Open problems

Several questions remain open and are explicitly out of the present paper’s scope.

- (O1) **Cokernel structure of $E_{\gamma,L}$.** The integer Smith normal form of $E_{\gamma,L}$ at the 22 odd-rank relabelings of **base1** produces five distinct diagonals ((15)); only 10/22 are torsion-free, the remaining 12 carry torsion in $\{2, 3, 4, 10\}$. A closed-form classification of which $\gamma \in \Gamma_{\text{base1}}^*$ produces which torsion class is open, as is the analogous distribution for M_{19} ’s 100 relabelings.
- (O2) **Hessian realization at $n = 25$.** Does the abstract group $5^2:Q_8$ arise as a Hessian symmetry of some Sudoku grid of order 25? At the abstract group level, the $m^2:Q_8$ family exists for every odd m . The Sudoku realization is verified at $m = 3$ in Theorem 8.5; it is open at $m = 5$.
- (O3) **Canonical-rank distribution at large n .** The sample histogram of rank $E_{\text{id},L}$ at $n = 9$ is $\{8:400\}$ on 400 Markov-walk samples. The high-corank tail $\{4, 6, 8\}$ predicted at small n by the rank-parity draft is currently unobserved at $n = 9$ in any sample of moderate size.
- (O4) **Trace–cokernel bridge.** A direct algebraic relation between $\text{tr}(\widehat{E}_{\gamma,L} \widehat{E}_{\gamma,L}^\top)$ and the cokernel structure is not known.
- (O5) **Universal dichotomy (Conjecture 9.3).** Whether the stabilizer dichotomy persists for every $n \geq 5$ remains open beyond the analytic ($n \in \{7, 9, 11\}$) and exhaustive ($5 \leq n \leq 13$) reach.

12.3 Out-of-scope, briefly

Two pointers, included as future-work signals only and *not* contributions of this paper.

The framework of §2 extends *verbatim* to several constrained Latin-square families: gerechte designs (general partition into equal-area boxes), rectangular Sudoku, orthogonal Sudoku Latin

squares, and constraint-satisfaction problems with local symmetry groups. None of these are studied here.

The Hessian group $3^2:Q_8$ has classical incarnations in the geometry of plane cubics, the projective A_8 root lattice, and in 4-dimensional polytope theory (the 600-cell stabilizer chain contains $2 \cdot 3^2:Q_8$ as a subgroup of low index). These geometric and topological motivations are deferred to companion projects and are not used in the present arguments.

A Exhaustive base₁ odd-rank list

The full list of the 22 odd-rank symbol relabelings $\gamma \in S_9$ on **base1** is contained in the field `outputs.odd_perms_full_list` of `base1_odd_rank_22.json`. Each entry is a permutation of $\{1, \dots, 9\}$ in one-line notation. By construction, the cardinality 22 matches Theorem 4.1; the four-way exact certification (rank, kernel, integer SNF, $\mathbb{F}_2$ SNF) is recorded in the same JSON under the `outputs.four_way_certificates` block. The certificate hash is pinned by [MANIFEST.sha256](#).

B M_{19} : 72 generators of K_L and presentation of $3^2:Q_8$

The Hessian group K_L at M_{19} is generated by the eight permutations of $\{1, \dots, 9\}$ recorded in the `outputs.K_L_generators_perms` field of `M19_KL_KR_3sq_Q8.json`. Their derived series of orders is $(72, 18, 9, 1)$, abelianization is $\mathbb{Z}/4$, exponent is 6, and the element-order multiset is $\{1:1, 2:21, 3:8, 4:18, 6:24\}$. As an abstract group, $K_L \cong 3^2 \rtimes Q_8 = \text{SmallGroup}(72, 41)$. The same data hold for K_R , which is conjugate to K_L via the structural permutation π_0 (Remark 8.6).

C Verified-range stabilizer table for $n \in \{5, \dots, 13\}$

For each n in $\{5, 6, 7, 8, 9, 10, 11, 12, 13\}$ the predicted and verified values of $|\text{Stab}(M_n)|$, $|\text{Stab}(O_n)|$, $|\text{orb}(M_n)|$, $|\text{orb}(O_n)|$, and the integer-hull elementary divisors of Λ_{M_n} and Λ_{O_n} are recorded in the `outputs.results` field of `stabilizers_5_to_13.json`. All nine rows pass `all_pass = true`. The total wall-time of the exhaustive verification is ≈ 31 s.

D Reproducibility manifest

The reproducibility package consists of:

- the eleven certified JSONs listed in §3 of [TIER_LEDGER.md](#);
- the SHA-256 manifest [MANIFEST.sha256](#);
- the verifier [verify_all.py](#) (claim-level invariant checks);
- the regenerator [reproduce_all.py](#) (byte-equal regeneration);
- the universality discipline document [UNIVERSALITY_LOCK.md](#);
- the sibling-notation reconciliation [SIBLING_NOTATION.md](#).

On the reference machine, `verify_all.py` returns 11/11 PASS in under 30 seconds, and `reproduce_all.py` reconstructs all eleven JSONs byte-equal to the manifest.

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